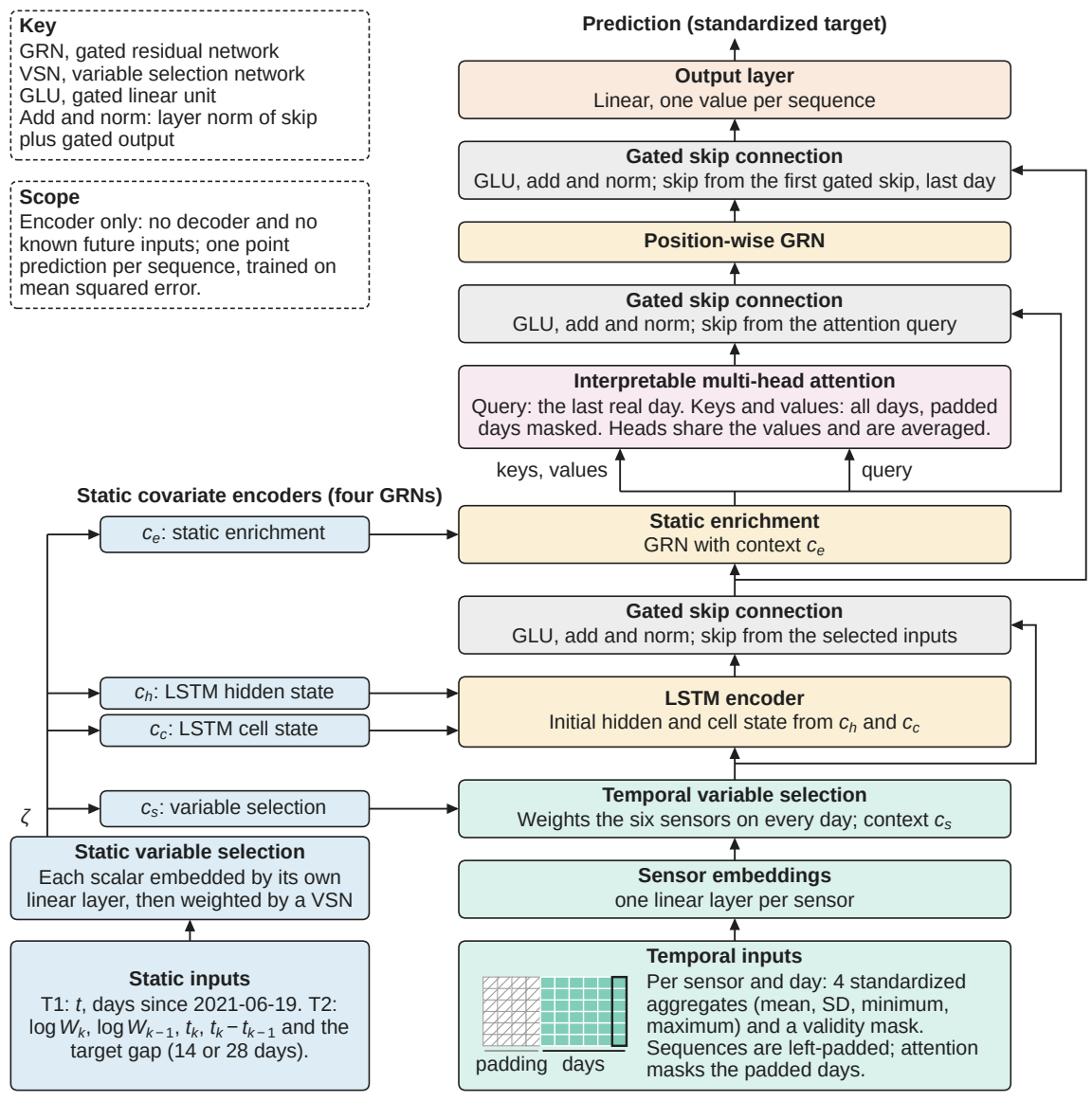
Key
GRN, gated residual network
VSN, variable selection network
GLU, gated linear unit
Add and norm: layer norm of skip plus gated output
Scope
Encoder only: no decoder and no known future inputs; one point prediction per sequence, trained on mean squared error.
Prediction (standardized target)
Output layer
Linear, one value per sequence
Gated skip connection
GLU, add and norm; skip from the first gated skip, last day
Position-wise GRN
Gated skip connection
GLU, add and norm; skip from the attention query
Interpretable multi-head attention
Query: the last real day. Keys and values: all days, padded days masked. Heads share the values and are averaged.
keys, values
query
Static covariate encoders (four GRNs)
c_e: static enrichment
Static enrichment
GRN with context c_e
Gated skip connection
GLU, add and norm; skip from the selected inputs
c_h: LSTM hidden state
c_c: LSTM cell state
LSTM encoder
Initial hidden and cell state from c_h and c_c
ζ
c_s: variable selection
Temporal variable selection
Weights the six sensors on every day; context c_s
Static variable selection
Each scalar embedded by its own linear layer, then weighted by a VSN
Sensor embeddings
one linear layer per sensor
Static inputs
T1: t, days since 2021-06-19. T2: $\log W_k$, $\log W_{k-1}$, t_k, $t_k - t_{k-1}$ and the target gap (14 or 28 days).
Temporal inputs
padding
days
Per sensor and day: 4 standardized aggregates (mean, SD, minimum, maximum) and a validity mask. Sequences are left-padded; attention masks the padded days.